

AICulpability_E15_AVDifferentLabels_AutoDrive_CoDrive (#252828)

Author(s)

This pre-registration is currently anonymous to enable blind peer-review.
It has 2 authors.

Pre-registered on:

2025/10/20 - 03:38 PM (PT)

1) Have any data been collected for this study already?

No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?

Our previous research indicated that marketing labels describing autonomous vehicles (AVs) can influence consumer perceptions of the vehicle's automation level. Furthermore, consumers are more likely to attribute moral responsibility to firms when they use marketing labels that inflate the perceived autonomy of vehicles. We found these effects when comparing within the label pairs "Autopilot vs Copilot" as well as "Full Self-Driving vs Lane Sense."

This study serves as an additional robustness check to generalize these findings to other conceptually similar labels: "AutoDrive" and "CoDrive." We predict that "AutoDrive" will be perceived as having a greater level of automation compared to "CoDrive," which will result in higher liability and responsibility attributions for the firm in the case of an accident.

3) Describe the key dependent variable(s) specifying how they will be measured.

The dependent variables in this study are:

1. What do you perceive as the level of automation for Aeon's [label] Program? (6-point scale from "Level 1 - not automated at all" to "Level 6 - fully automated")
2. Aeon's [label] is responsible for the accident. (100-point scale from 0 = "Strongly Disagree" to 100 = "Strongly Agree")
3. The human in the car is responsible for the accident. (100-point scale from 0 = "Strongly Disagree" to 100 = "Strongly Agree")
4. Aeon, the company, is liable for the damages from the accident. (100-point scale from 0 = "Strongly Disagree" to 100 = "Strongly Agree")
5. The human in the car is liable for the damages from the accident. (100-point scale from 0 = "Strongly Disagree" to 100 = "Strongly Agree")

4) How many and which conditions will participants be assigned to?

There will be two conditions (label: 'AutoDrive' vs. 'CoDrive'). Participants will be randomly assigned to one condition in a between-subjects design.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

If the measures of human responsibility and liability (DVs 3 and 5) have a Cronbach's alpha of 0.7 or higher, we will average these measures to form a "human liability" composite. The same procedure will be applied to the measures of firm/software responsibility and liability (DVs 2 and 4) to form a "firm liability" composite.

We will conduct t-tests to compare the differences between the 'AutoDrive' and 'CoDrive' conditions for all dependent variables (and composites). Additionally, we will perform mediation analyses where the dependent variables are human liability and firm liability (or their composites), the mediator is the perceived level of automation, and the independent variable is the label condition.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

We will exclude participants who answer any one of three comprehension check questions incorrectly.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

We will determine our sample size using a Pre-Registered Interim Analysis Design (PRIAD), specifically the two-stage Pocock design outlined by Andre and Reinholtz (2024). An initial sample of 500 participants will be collected. We will stop data collection at this stage if the key measures and mediation effect yield a p-value below 0.0294. Otherwise, we will continue data collection to achieve a full, final sample of 1,000 participants.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

Only participants who pass at least two attention checks at the beginning of the survey will be eligible. We will include non-identifiable demographic questions (age, gender). We will also ask participants how familiar they are with AVs on a 100-point scale (0 = Very little to 100 = A lot), which will be included as a covariate in exploratory robustness checks.